Milestone 1: Social Returns of The Harbour, Victoria B.C.'s Supervised Consumption Site

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2026-10-20

Victoria B.C.'s safe consumption site, The Harbour, is located on the city's infamous 900-block of Pandora Avenue. Operated by Island Health and SOLID Outreach, the facility is home to North America's first indoor group inhalation site and allows injection as well. The dataset used in this paper was collected as a part of a study funded by the British Columbia Ministry of Health and led by Dr. Michael Ma of the University of Victoria. This paper features simulated data for a select number of the dataset's response categories which have been selected to answer this paper's research question.

Introduction: Description of Dataset and Research Question

As a student researcher on a Ministry of Health funded project led by Dr. Michael Ma (UVic) for over a year, I chose to use the survey data we collected as a part of this mixed methods study and further analyze it using R for my SOCI 502 Data Story project. Members (users) of The Harbour are housed or unhoused, use various types of drugs, range in age, race and gender, spend varying amounts of time at the site and for various purposes. The question I am asking is as follows: How is the time members spend at The Harbour associated with participation in non-consumption activities, and does this association vary with travel time to the site and drug use patterns?

To answer the research question, descriptive statistics and graphs will be used to determine whether the mean and median time spent differs for those respondents who partake in non-consumption activity (in addition to or in the absence of consumption activity) at The Harbour compared to those who only visit the site to consume illicit substances. To address the second part of the question, these differences will be tested against those whose travel time to the site is short, medium or long. While it is difficult to estimate without the use of probability tests, a difference of

Loading Packages and Setting Up the Workspace

In order to simulate the data, the packages "tidyverse" and "janitor" need to be loaded. Given that they have previously been installed onto my machine, they now only need to be loaded.

```
library(tidyverse)
library(janitor)
library(extraDistr)
library(dplyr)
library(tinytable)
```

Simulating Variables

In order to answer the above question, I have chosen to use various response categories/survey questions from the dataset. The most important ones are "TimeAtHarb", "TimeToHarb", "CometoInject", "CometoSmoke," "CometoGetSubstances", "CometoDrugCheck", "CometoSeeFriends", "CometoSeeStaff", "ComeforODPrevention", "ComeforCleanSupplies", "ComeforSafety", "ComeforPrivacy", "ComeforWarmthShelter", "CometoChargePhone", "CometoBathroom", "ComeforCounselling", "ComeforDealers", "ComeforStreetNurse", "ComeforOtherServices".

The "TimeAtHarb" variable used response categories ranging from 1-8, for "TimeToHarb", responses varied (save some outliers) from 0-60 minutes, and each response to "Why do you come to The Harbour?" are in binary (1 = reason for coming, 0 = not a reason). These measures are taken into account when simulating the variables below. I have chosen to leave the outliers out of the data simulation as it's purpose is to ensure the code works rather than provide accurate data.

```
library(dplyr)

set.seed(1836)
n_records <- 212

# Defining the 17 binary variable names.
var_names <- c(
  "CometoInject", "CometoSmoke", "CometoGetSubstances", "CometoDrugCheck",
  "CometoSeeFriends", "CometoSeeStaff", "ComeforODPrevention", "ComeforCleanSupplies",
  "ComeforSafety", "ComeforPrivacy", "ComeforWarmthShelter", "CometoChargePhone",
  "CometoBathroom", "ComeforCounselling", "ComeforDealers", "ComeforStreetNurse",
  "ComeforOtherServices"
)
```

```r
# Defining probabilities for each variable (all set to 0.50).
probabilities <- rep(0.50, length(var_names))

# Simulating the matrix for the binary variables.
sim_matrix <- matrix(
  rbinom(n_records * length(var_names), size = 1, prob = probabilities),
  nrow = n_records,
  ncol = length(var_names),
  byrow = TRUE
)

# Converting to a data frame and adding column names.
visit_data <- as.data.frame(sim_matrix)
colnames(visit_data) <- var_names


# Now, adding the other two variables and adding the matrix to them in a tibble.
simulated_Harbour_data <- tibble(
  "Participant" = 1:n_records,
  "TimeAtHarb" = sample(
    x = c(1:8),
    size = n_records,
    replace = TRUE),
  "TimeToHarb" = sample(
    x = c(0:60),
    size = n_records,
    replace = TRUE)
) |>
  bind_cols(visit_data) # This attaches all 17 binary columns at once.

# Viewing the resulting simulated data frame.
#| tbl-pos: H
simulated_Harbour_data |>
  head(10)
```

```
# A tibble: 10 x 20
   Participant TimeAtHarb TimeToHarb CometoInject CometoSmoke
         <int>      <int>      <int>        <int>       <int>
 1           1          7         26            0           0
 2           2          2         13            0           0
 3           3          1         29            0           1
```

```
 4           4           6          30           1           0
 5           5           1          10           0           1
 6           6           3          14           0           0
 7           7           5          20           0           1
 8           8           1          12           1           0
 9           9           3          47           1           1
10          10           4          12           1           1
# i 15 more variables: CometoGetSubstances <int>, CometoDrugCheck <int>,
#   CometoSeeFriends <int>, CometoSeeStaff <int>, ComeforODPrevention <int>,
#   ComeforCleanSupplies <int>, ComeforSafety <int>, ComeforPrivacy <int>,
#   ComeforWarmthShelter <int>, CometoChargePhone <int>, CometoBathroom <int>,
#   ComeforCounselling <int>, ComeforDealers <int>, ComeforStreetNurse <int>,
#   ComeforOtherServices <int>
```

```
# Now, I am making it into a table.
#| tbl-pos: H
simulated_Harbour_data |>
  select(TimeAtHarb, TimeToHarb, 4:8) |>
  head(5) |>
  t() |>
  as.data.frame() |>
  tibble::rownames_to_column("Variable") |>
  tt() |>
  setNames(c("Variable", paste("Participant", 1:5)))
```

Variable	Participant 1	Participant 2	Participant 3	Participant 4	Participant 5
TimeAtHarb	7	2	1	6	1
TimeToHarb	26	13	29	30	10
CometoInject	0	0	0	1	0
CometoSmoke	0	0	1	0	1
CometoGetSubstances	1	1	0	0	1
CometoDrugCheck	0	0	0	0	1
CometoSeeFriends	0	0	1	1	0

```
# Alternatively, this gives the sums of the reason columns.
#| tbl-pos: H
tibble(Reason = var_names,
       n = colSums(simulated_Harbour_data[, var_names])) |>
  tt()
```

Reason	n
CometoInject	106
CometoSmoke	100
CometoGetSubstances	107
CometoDrugCheck	93
CometoSeeFriends	111
CometoSeeStaff	80
ComeforODPrevention	114
ComeforCleanSupplies	106
ComeforSafety	101
ComeforPrivacy	109
ComeforWarmthShelter	96
CometoChargePhone	110
CometoBathroom	102
ComeforCounselling	90
ComeforDealers	101
ComeforStreetNurse	103
ComeforOtherServices	111

References

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